Introduction to

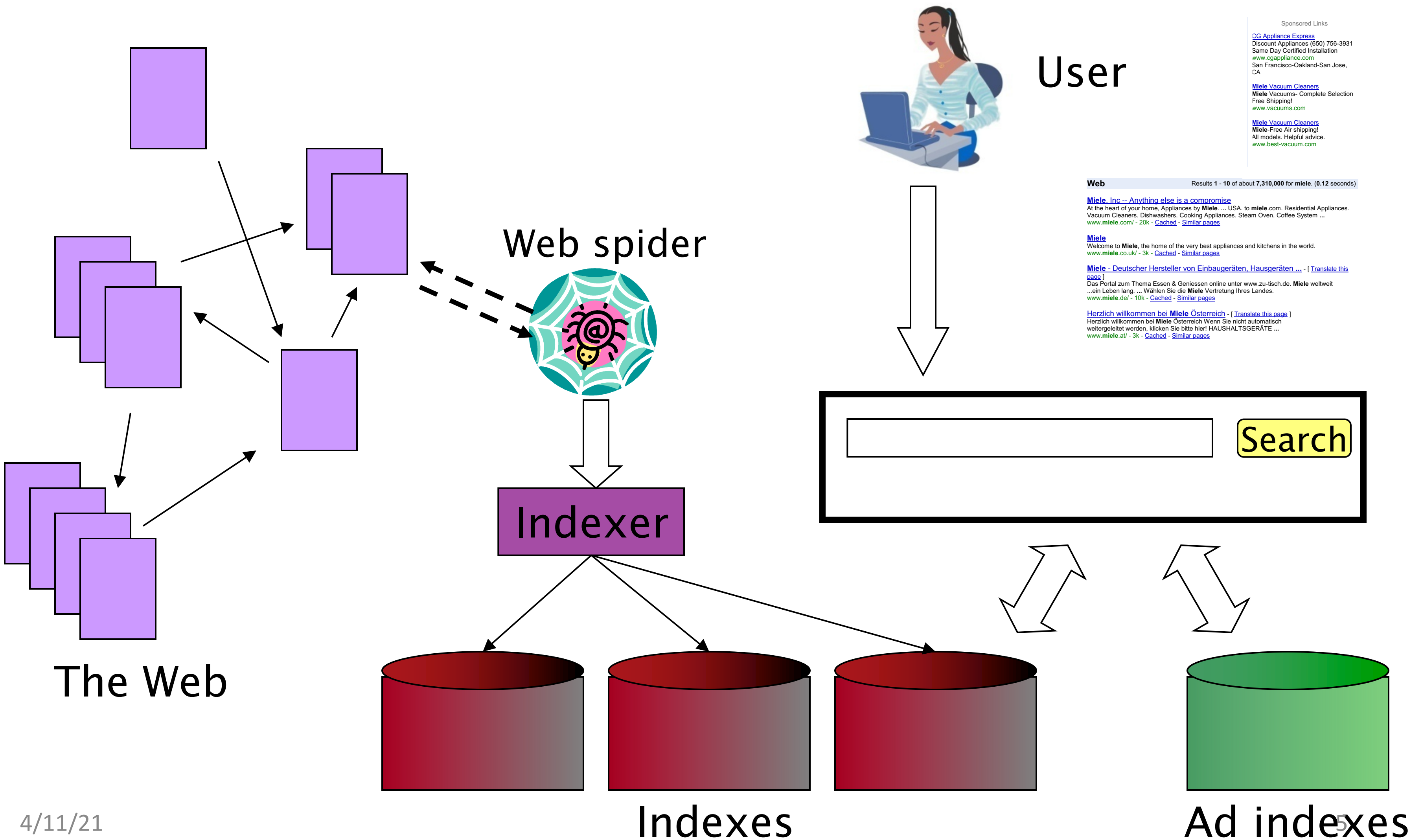
Information Retrieval

Review

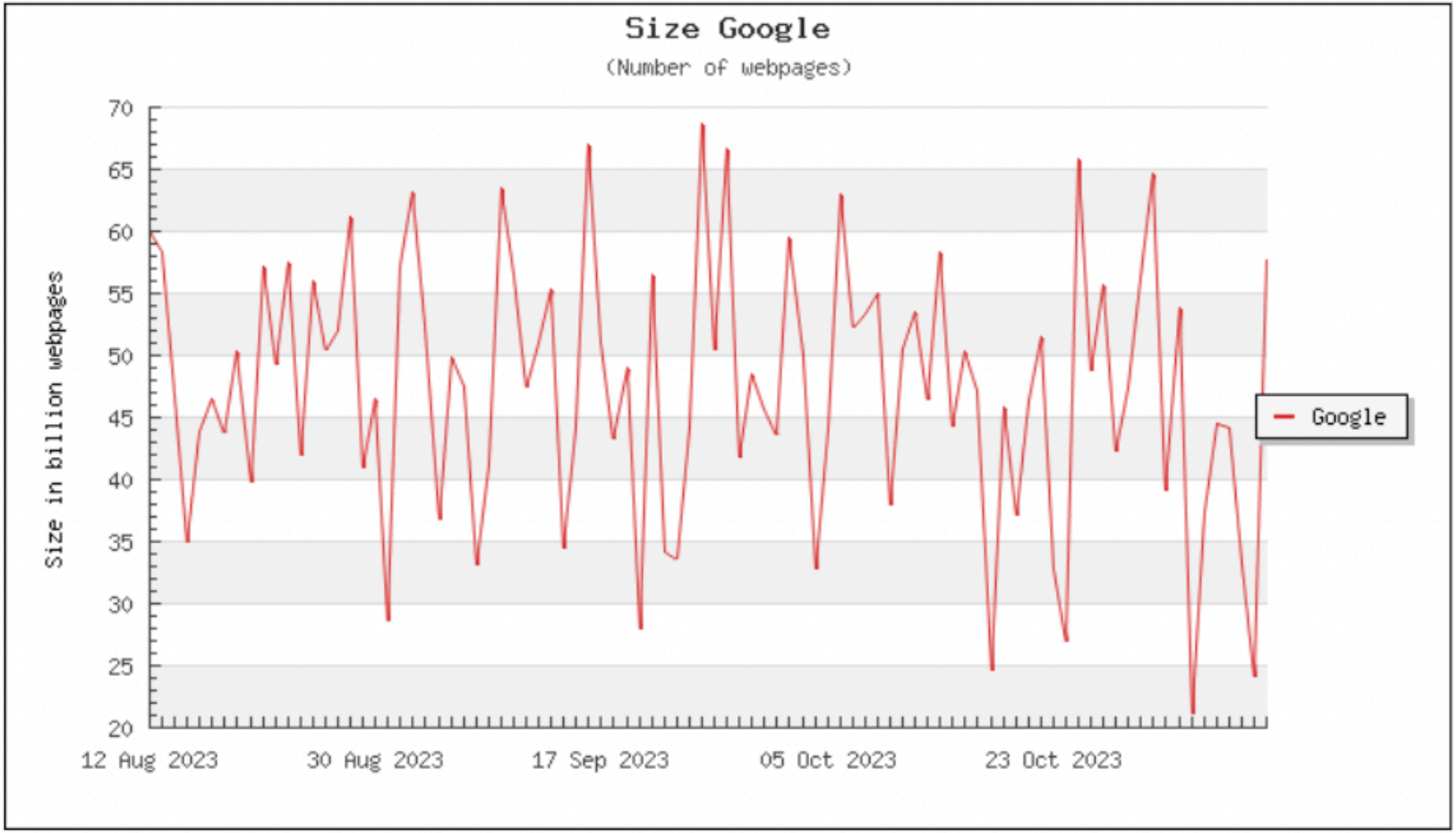
Agenda

- Project
- Lecture revisions
- Final exam
- Announcements

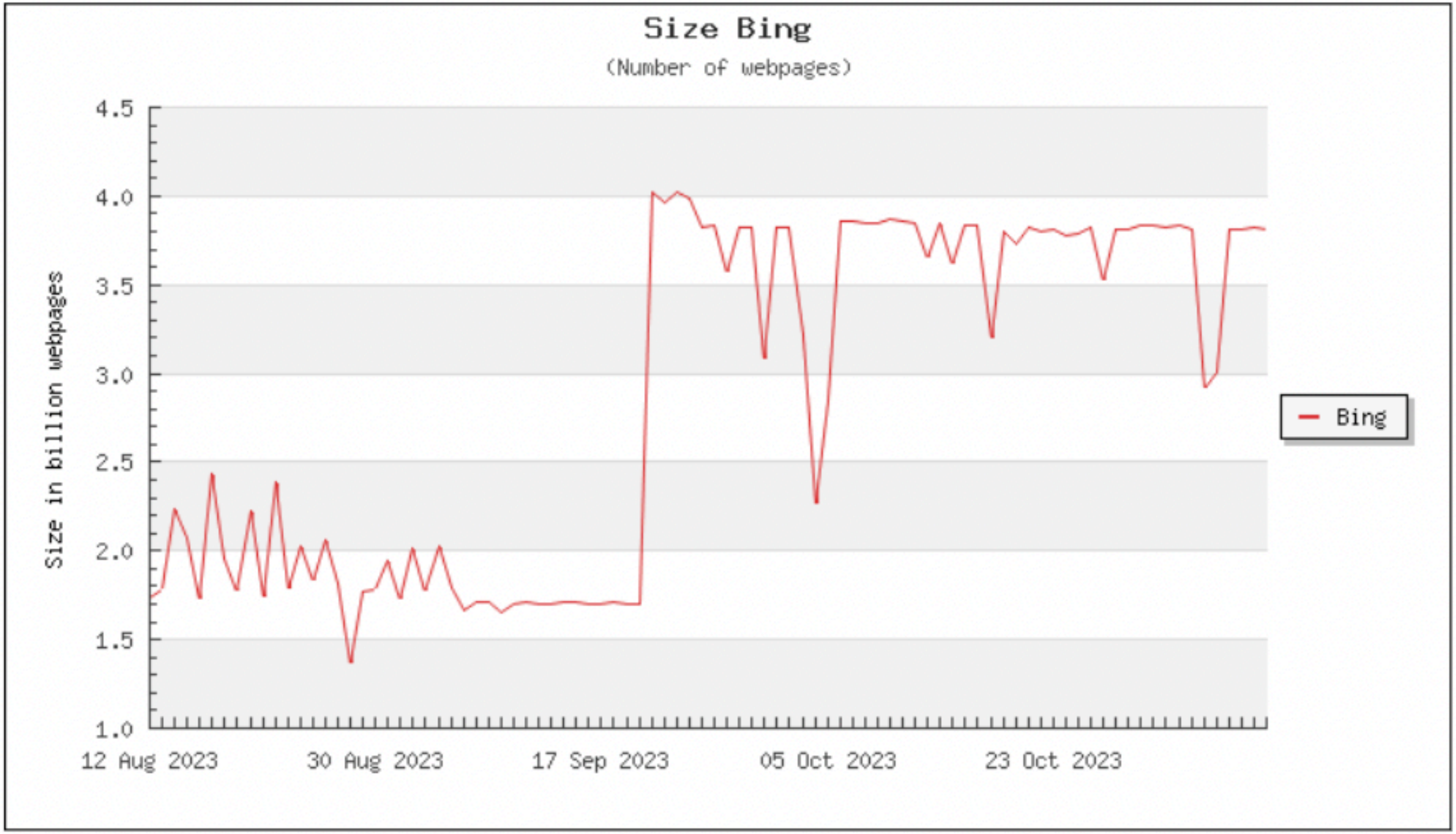
Web search basics



Project related

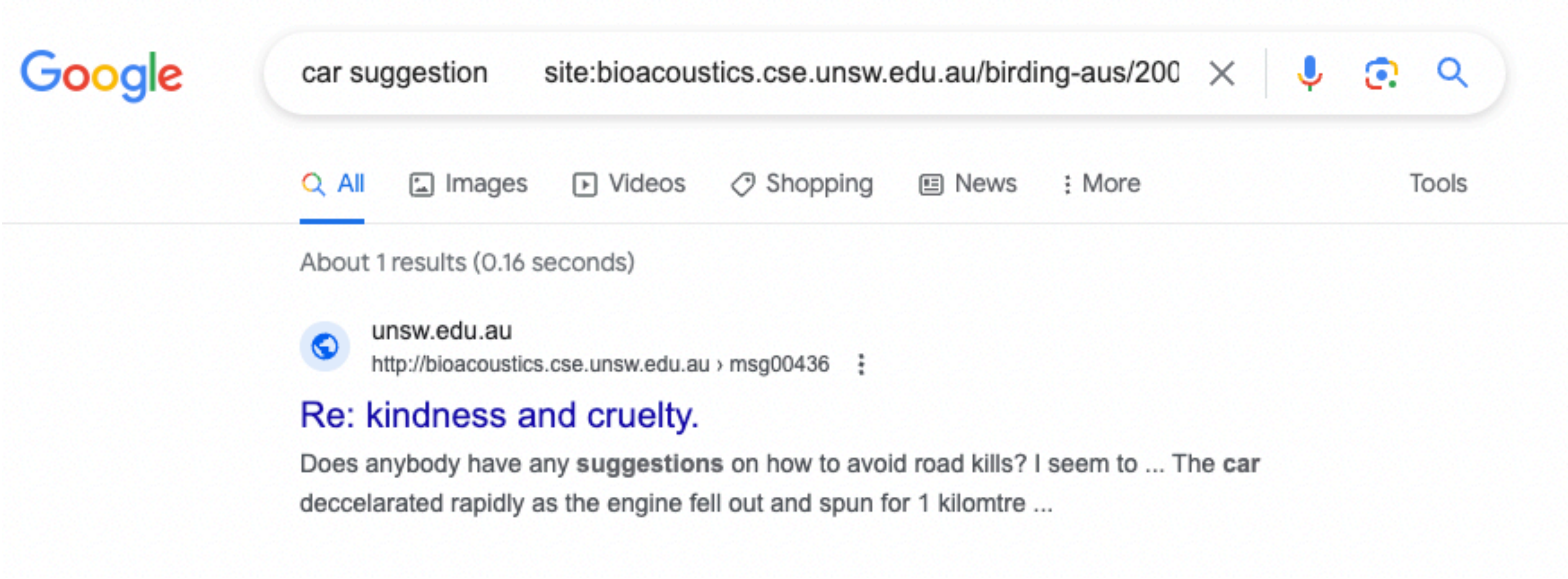


Project related

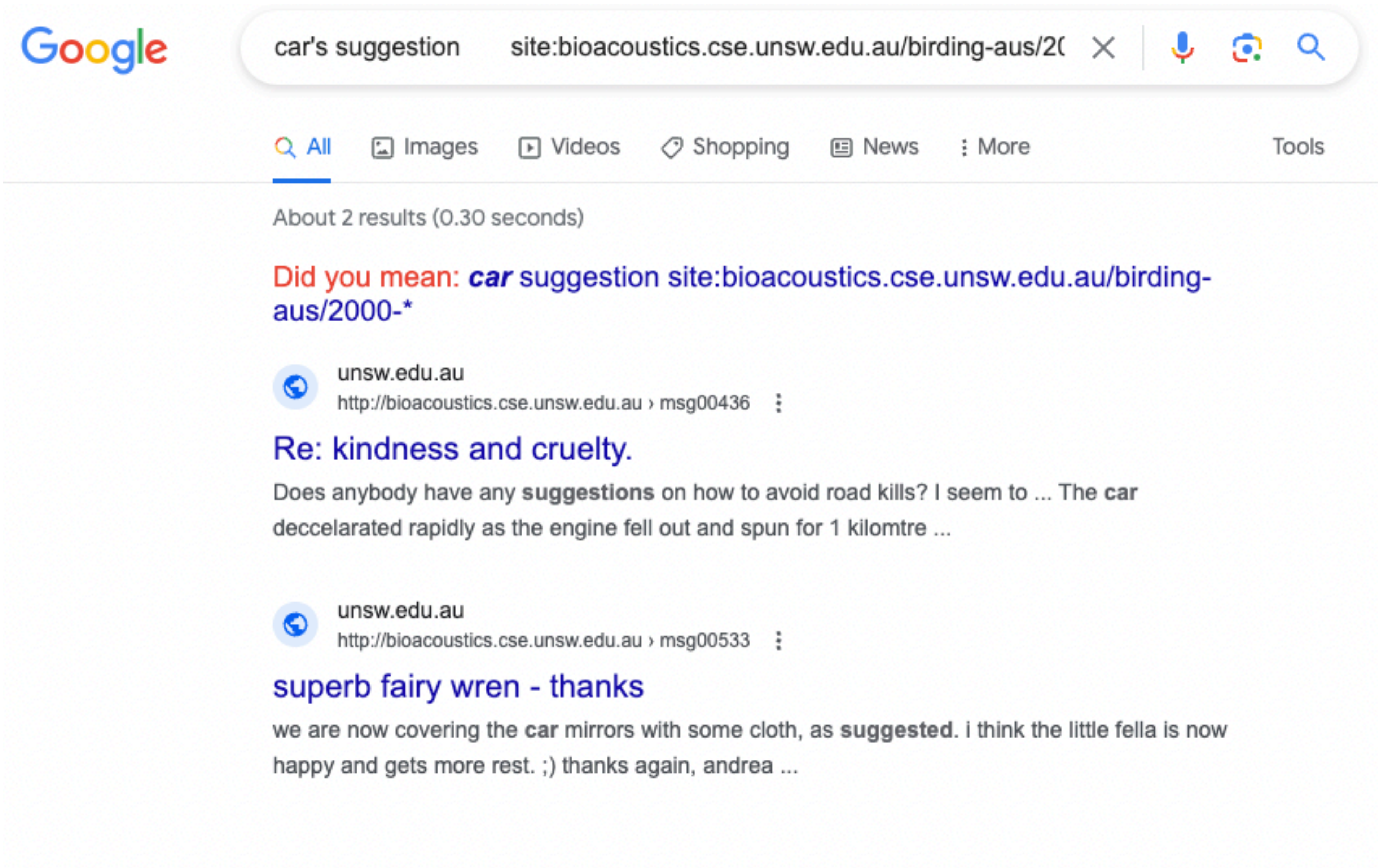


Project related

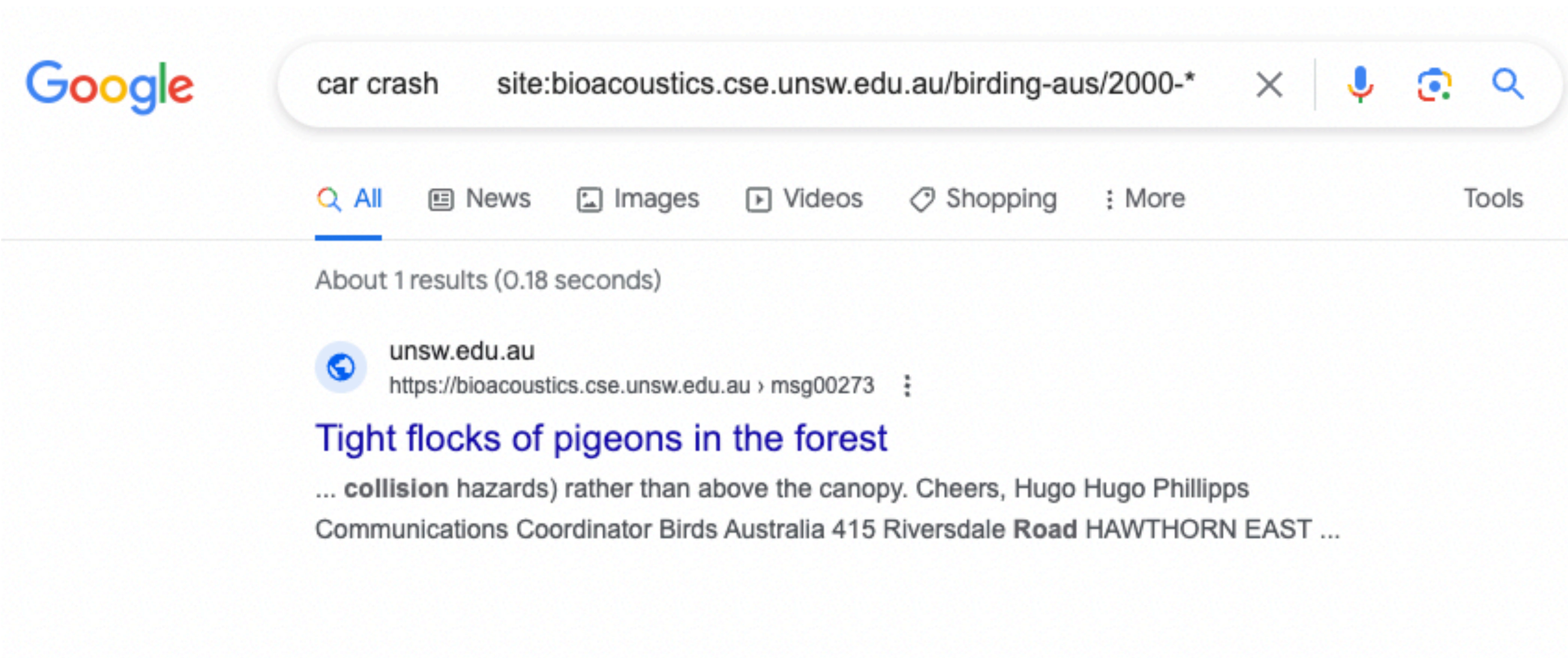
Project related



Project related (issue)



Project related (issue)



Lecture revisions

Boolean Model

- incidence vector/matrix
- semantics of the query model (AND/OR/NOT, and other operators, e.g., /k, /S)
- inverted index, positional inverted index
- query processing methods for basic and advanced boolean queries (including phrase query, queries with /S operator, etc.)
- query optimization methods (list merge order, skip pointers)

Preprocessing

- typical preprocessing steps: tokenization, stopword removal, stemming/lemmatization,

Tolerant Retrieval

- Wildcard queries
- Permuterm index
- Bigram index
- Spelling correction
- Noise channel model

Other Topics

- Soundex
- N-grams
- Maximum Likelihood Estimate

Index Construction

- Why we need dedicated algorithms to build the index?
- BSBI: Blocked sort-based indexing
- SPIMI: Single-pass in-memory indexing
- Dynamic indexing: Immediate merge, no merge, logarithmic merge

Index Compression

- Heap’s law, Zipf’s law
- Dictionary compression
  - Dictionary as a string
  - Front encoding
- posting lists compression
  - Elias encoding
  - Variable length encoding
- Shannon limit

Vector Space Model

- What is/why ranked retrieval?
- raw and normalized tf, idf
- cosine similarity
- tf-idf variants (using SMART notation): e.g., lnc.ltc
- basic query processing method: document-at-a-time vs term-at-a-time
- exact & approximate query optimization methods (heap-based top-k algorithm, MaxScore algorithms, etc.)
- Query processing methods based on advanced or tiered inverted indexes (e.g., high/low lists)

tf-idf example: lnc.ltc

Document: *car insurance auto insurance*
Query: *best car insurance*

Term	Query						Document				Prod
	tf-raw	tf-wt	df	idf	wt	n'lize	tf-raw	tf-wt	wt	n'lize	
auto	0	0	5000	2.3	0	0	1	1	1	0.52	0
best	1	1	50000	1.3	1.3	0.34	0	0	0	0	0
car	1	1	10000	2.0	2.0	0.52	1	1	1	0.52	0.27
insurance	1	1	1000	3.0	3.0	0.78	2	1.3	1.3	0.68	0.53

Exercise: what is *N*, the number of docs?

Doc length $= \sqrt{1^2 + 0^2 + 1^2 + 1.3^2} \approx 1.92$

Score = 0+0+0.27+0.53 = 0.8

Cosine for length-normalized vectors

- For length-normalized vectors, cosine similarity is simply the dot product (or scalar product):

$$\cos(\vec{q},\vec{d}) = \vec{q} \bullet \vec{d} = \sum_{i=1}^{|V|} q_i d_i$$

for q, d length-normalized.

Evaluation

- Existing method to prepare for the benchmark dataset, queries, and ground truth
- For unranked results: Precision, recall, F-measure
- For ranked results: precision-recall graph, 11-point interpolated
- precision, MAP, etc.

Web Search Basics

- Estimation of relative sizes of two search engines.
- Near duplicate detection: the shingling method

Crawling

- Understand the requirements of crawlers
- Mercator scheme
- **Not required**: optimization for age

Link Analysis

- The pagerank algorithm: theory and practice

COMP6714 vs COMP9319

- COMP6714 – Information Retrieval
  - Search basics, search quality
  - Python
- COMP9319 – Web Data Compression and Search
  - Size, performance & scalability
  - C/C++

Week	Topic
1	Course Introduction + Boolean Retrieval
2	Preprocessing
3	Index construction
4	Compression
5	Vector Space Model
6	Flexibility Week (no lecture)
7	Evaluation
8	Crawling
9	Link Analysis
10	Revision and Exam Preparation

Assessment

```
asgt        = mark for the assignment    (out of 10)
proj        = mark for the project       (out of 40)
exam        = mark for final exam        (out of 50)
okEach      = exam > 20                  (after scaling)
mark        = asgt + proj + exam
grade       = HD|DN|CR|PS  if mark >= 50 && okEach
            = FL           if mark <  50 && okEach
            = UF           if !okEach
```

Final exam (2hrs, worth 50%)

- Exam date: In-person, 30 Nov (Thu) afternoon, 1:45pm-4:00pm
- Exam instructions will be available 15mins before 2:00pm
- If you are ill on the day of the exam, do not attend the exam – c.f. UNSW fit-to-sit policy. Apply for special consideration asap.
- If granted by UNSW SC, supp exam is likely to be of different format, e.g., long questions / programming / oral exam / mixed

Final exam format

- 2 hours exam, closed book exam
- You may refer to the lecture slides
- Bring a **UNSW-approved calculator** & **pens**
- A programming question + M.C. / written questions
- Similar "style" as your assignment but of a larger scope

Access to Python documentation

Python 3.9.17 documentation

Welcome! This is the official documentation for Python 3.9.17.

Parts of the documentation:

What's new in Python 3.9?
or all "What's new" documents since 2.0

Tutorial
start here

Library Reference
keep this under your pillow

Language Reference
describes syntax and language elements

Python Setup and Usage
how to use Python on different platforms

Python HOWTOs
in-depth documents on specific topics

Installing Python Modules
installing from the Python Package Index & other sources

Distributing Python Modules
publishing modules for installation by others

Extending and Embedding
tutorial for C/C++ programmers

Python/C API
reference for C/C++ programmers

FAQs
frequently asked questions (with answers!)

Indices and tables:

Global Module Index
quick access to all modules

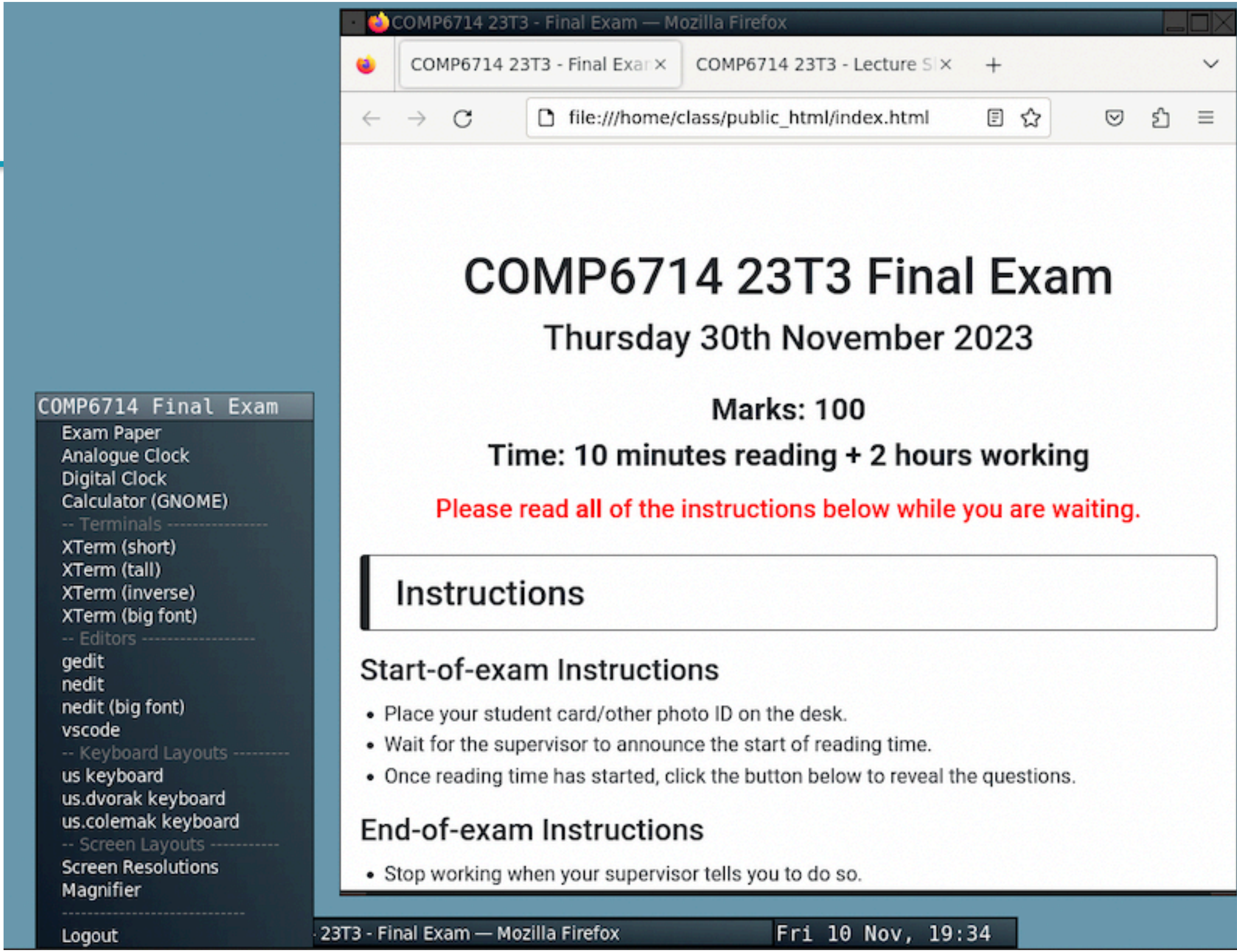
General Index
all functions, classes, terms

Glossary

Search page
search this documentation

Complete Table of Contents
lists all sections and subsections

How to prepare for the exam…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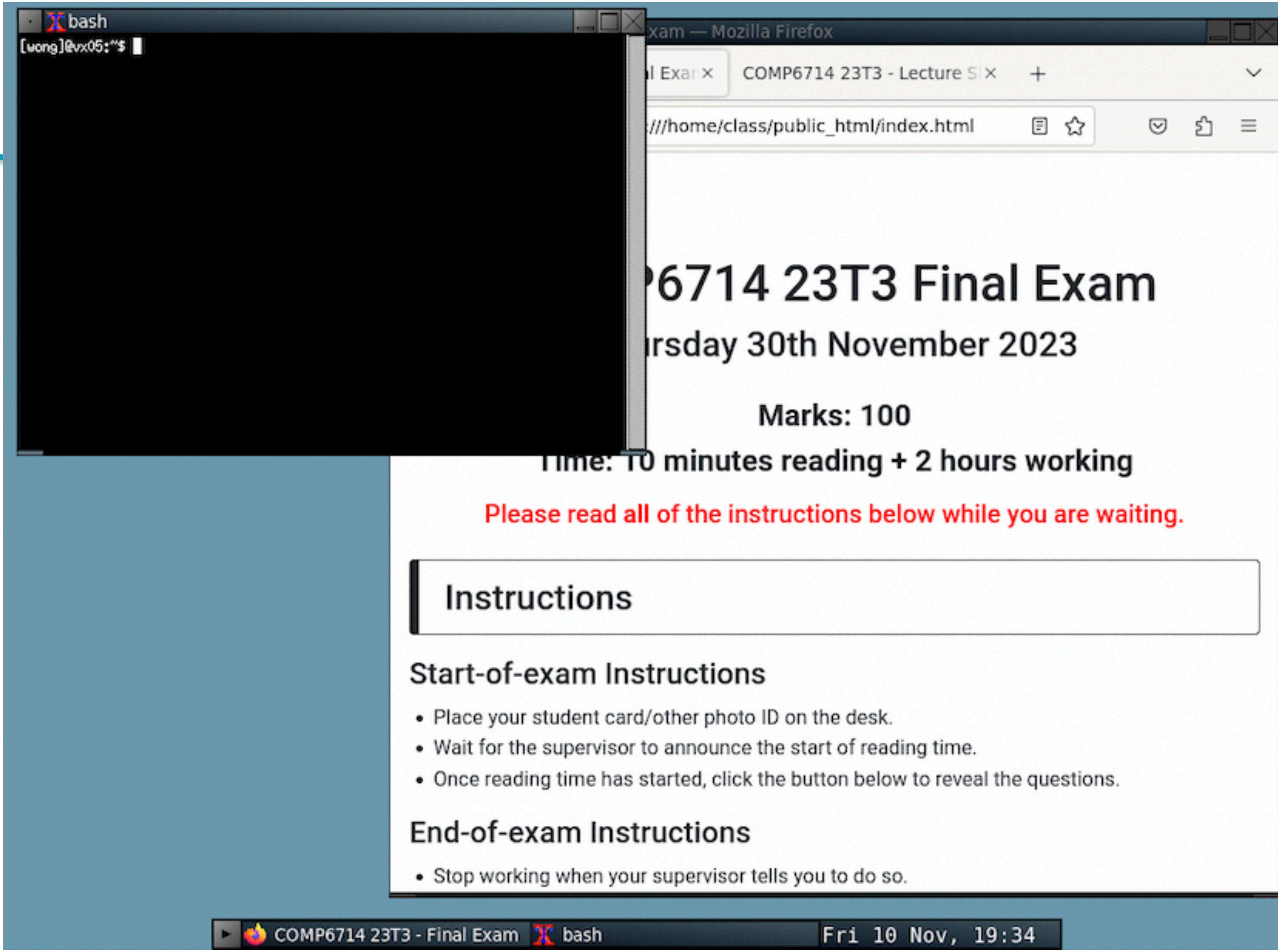
COMP6714 23T3 - Final Exar
COMP6714 23T3 - Lecture S
file:///home/class/public_html/index.html
COMP6714 23T3 Final Exam
Thursday 30th November 2023
Marks: 100
Time: 10 minutes reading + 2 hours working
Please read all of the instructions below while you are waiting.
Instructions
Start-of-exam Instructions
• Place your student card/other photo ID on the desk.
• Wait for the supervisor to announce the start of reading time.
• Once reading time has started, click the button below to reveal the questions.
End-of-exam Instructions
• Stop working when your supervisor tells you to do so.
COMP6714 Final Exam
Exam Paper
Analogue Clock
Digital Clock
Calculator (GNOME)
-- Terminals ---------------
XTerm (short)
XTerm (tall)
XTerm (inverse)
XTerm (big font)
-- Editors ----------------
gedit
nedit
nedit (big font)
vscode
-- Keyboard Layouts ---------
us keyboard
us.dvorak keyboard
us.colemak keyboard
-- Screen Layouts -----------
Screen Resolutions
Magnifier
Logout
23T3 - Final Exam — Mozilla Firefox
Fri 10 Nov, 19:34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bash
Fri 10 Nov, 19:34

A sample question

Question 7 (3 marks)

*Write your answers for this question in **q7.txt**.*

Which of the following are queries that skip pointers are useful for?

A	x OR y
B	x AND y
C	x OR (NOT y)
D	x AND (NOT y)
E	NOT x

Submission

Submit using the following command:

```
$ submit q7
```

q7.txt

```
q7.txt

Question 7

Please replace "X, Y, Z" inside the square brackets with your
answer, e.g., "A, B". Do not remove the square brackets.


### ANSWER: [ X, Y, Z ]
```

q7.txt

```
q7.txt

Question 7

Please replace "X, Y, Z" inside the square brackets with your
answer, e.g., "A, B". Do not remove the square brackets.


### ANSWER: [ B, D ]
```

Another sample question

Question 8 (6 marks)

*Write your answers for this question in **q8.txt**.*

Suppose a search engine has indexed totally 10 documents, with 3 of them containing a keyword Apple.

A search on Apple returns 5 documents. Of the 5 documents retrieved, 3 contains Apple. Its results are as follows (the leftmost item is the top ranked search result):

A A N A N

where A and N represent documents containing Apple and those without Apple respectively.

Based on the above query, what is the MAP (mean average precision) of the search engine?

Round your answer to two decimal places.

Submission

Submit using the following command:

```
$ submit q8
```

q8.txt

q8.txt ×

Question 8

Please replace "X" inside the square brackets with your answer. Do not remove the square brackets.

ANSWER: [X]

q8.txt

```
q8.txt

Question 8

Please replace "X" inside the square brackets with your
answer. Do not remove the square brackets.


### ANSWER: [ 0.92 ]
```

Typical (good) 11 point precisions

- SabIR/Cornell 8A1 11pt precision from TREC 8 (1999)

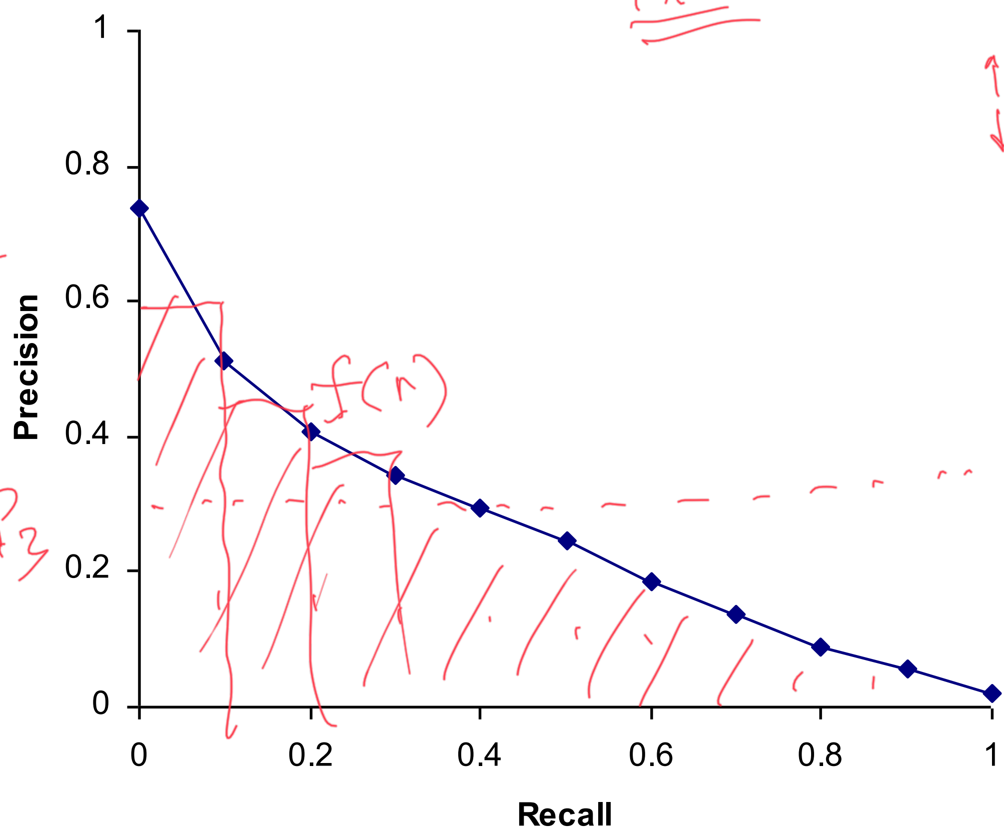
Q1 AP1

Q2 AP2

Q3 AP3

$\frac{\sum_{i}^{n} AP_i}{n}$

AP



Area

$h = \frac{Area}{b-a}$

$h = \frac{1}{b-a}\int f$

Q8

Table 1

	A	B	C	D	E	F
1						
2		prec@1	prec@2	prec@3	prec@4	prec@5
3		1	1	0.67	0.75	0.6
4	**Recall**	0.33	0.67	0.67	1	1
5						
6						
7		0.916666666666667				
8						
9						

Formula (B3 + C3 + E3)÷3

Yet more evaluation measures…

- Mean average precision (MAP)
  - Average of the precision value obtained for the top *k* documents, each time a relevant doc is retrieved
  - Avoids interpolation, use of fixed recall levels
  - MAP for query collection is arithmetic ave.
    - Macro-averaging: each query counts equally
- R-precision
  - If have known (though perhaps incomplete) set of relevant documents of size *Rel,* then calculate precision of top *Rel* docs returned
  - Perfect system could score 1.0.

Announcements

- Consultations (Monday in-person, Thursday online) will run till the final exam.
- Ed forum will be monitored and answered till the final exam.
- Project marking starts shortly (aim to finish by the end of next week) - you will be notified.

Finally

MyExperience: due very soon

Please give some constructive feedback, **thank you** !

Thanks for taking COMP6714

Good luck for your exam